

EXERCISE 8.3

Geometric Progressions

Questions with Step-by-Step Solutions

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Key Idea

A sequence is called a **Geometric Progression (GP)** if the ratio of consecutive terms is always the same.

This fixed number is called the **common ratio** r .

$$r = \frac{\text{next term}}{\text{previous term}}$$

If the first term is a and common ratio is r , then

$$t_n = ar^{n-1}$$

Question 1

Find the 12th term of a GP whose common ratio is 2 and whose 8th term is 192.

Solution

Given,

$$r = 2, \quad t_8 = 192$$

For a GP,

$$t_n = ar^{n-1}$$

For the 8th term,

$$\begin{aligned} t_8 &= ar^7 \\ 192 &= a(2)^7 \\ 192 &= 128a \\ a &= \frac{192}{128} \\ a &= \frac{3}{2}. \end{aligned}$$

Now find the 12th term:

$$\begin{aligned} t_{12} &= \frac{3}{2}(2)^{11} \\ &= 3(2)^{10} \\ &= 3(1024) \\ &= \boxed{3072}. \end{aligned}$$

Final Answer

$$t_{12} = \boxed{3072}$$

Question 2

Find the 10th and the n th terms of the GP

$$5, 25, 125, \dots$$

Solution

The first term is

$$a = 5$$

The common ratio is

$$\begin{aligned} r &= \frac{25}{5} \\ &= 5. \end{aligned}$$

Using

$$t_n = ar^{n-1},$$

we get

$$\begin{aligned} t_n &= 5(5)^{n-1} \\ &= \boxed{5^n}. \end{aligned}$$

For the 10th term,

$$\begin{aligned} t_{10} &= 5^{10} \\ &= \boxed{9765625}. \end{aligned}$$

Final Answer

$$t_{10} = 9765625$$

$$t_n = 5^n$$

Question 3

A sequence is defined recursively by

$$t_1 = 2, \quad t_{n+1} = 3t_n - 2, \quad n \geq 1.$$

Which term of the sequence is 730?

Solution

We calculate the terms one by one.

$$t_1 = 2,$$

$$\begin{aligned} t_2 &= 3t_1 - 2 \\ &= 3(2) - 2 \\ &= 4, \end{aligned}$$

$$\begin{aligned} t_3 &= 3t_2 - 2 \\ &= 3(4) - 2 \\ &= 10, \end{aligned}$$

$$\begin{aligned} t_4 &= 3t_3 - 2 \\ &= 3(10) - 2 \\ &= 28, \end{aligned}$$

$$\begin{aligned} t_5 &= 3t_4 - 2 \\ &= 3(28) - 2 \\ &= 82, \end{aligned}$$

$$\begin{aligned} t_6 &= 3t_5 - 2 \\ &= 3(82) - 2 \\ &= 244, \end{aligned}$$

$$\begin{aligned} t_7 &= 3t_6 - 2 \\ &= 3(244) - 2 \\ &= 730. \end{aligned}$$

Therefore,

$$\boxed{730 = t_7}$$

Final Answer

$\boxed{730 \text{ is the 7th term.}}$

Question 4

Which term of the GP

$$2, 6, 18, \dots$$

is 4374?

Also write the explicit formula and the recursive formula for the n th term.

Solution

The first term is

$$a = 2$$

and the common ratio is

$$\begin{aligned} r &= \frac{6}{2} \\ &= 3. \end{aligned}$$

Therefore,

$$t_n = 2(3)^{n-1}.$$

We need

$$t_n = 4374.$$

Hence,

$$\begin{aligned} 2(3)^{n-1} &= 4374 \\ 3^{n-1} &= 2187 \\ 3^{n-1} &= 3^7. \end{aligned}$$

Therefore,

$$\begin{aligned} n - 1 &= 7 \\ n &= 8. \end{aligned}$$

Thus, 4374 is the 8th term.

Explicit formula:

$$t_n = 2(3)^{n-1}$$

Recursive formula:

$$t_1 = 2, \quad t_{n+1} = 3t_n$$

Final Answer

4374 is the 8th term

$$t_n = 2(3)^{n-1}$$

$$t_1 = 2, \quad t_{n+1} = 3t_n$$

Question 5

A ball is dropped from a height of 80 metres. After hitting the ground, it bounces back to 60% of the height from which it fell. It continues bouncing in the same way.

- (i) What height does the ball reach after the 5th bounce?
- (ii) What total vertical distance has the ball travelled by the time it hits the ground for the 6th time?

Solution

The ball rises to 60% of its previous height.

Therefore,

$$r = 60 \frac{100}{100} = 0.6.$$

The heights reached after successive bounces are

$$48, 28.8, 17.28, 10.368, 6.2208, \dots$$

This is a GP with

$$a = 48, \quad r = 0.6.$$

(i) Height after the 5th bounce

Using

$$t_n = ar^{n-1},$$

we get

$$\begin{aligned} t_5 &= 48(0.6)^4 \\ &= 48(0.1296) \\ &= \boxed{6.2208 \text{ m}}. \end{aligned}$$

(ii) Total distance travelled until the 6th ground hit

The ball first falls through 80 m.

After that, every bounce height is travelled twice: once upward and once downward.

Therefore,

$$\begin{aligned} D &= 80 + 2(48 + 28.8 + 17.28 + 10.368 + 6.2208) \\ &= 80 + 2(110.6688) \\ &= 80 + 221.3376 \\ &= \boxed{301.3376 \text{ m}}. \end{aligned}$$

Final Answer

$$\boxed{\text{Height after 5th bounce} = 6.2208 \text{ m}}$$

$$\boxed{\text{Total distance} = 301.3376 \text{ m}}$$

Question 6

Which term of the sequence

$$2, 2\sqrt{2}, 4, \dots$$

is 128?

Solution

The first term is

$$a = 2.$$

The common ratio is

$$\begin{aligned} r &= \frac{2\sqrt{2}}{2} \\ &= \sqrt{2}. \end{aligned}$$

Therefore,

$$t_n = 2(\sqrt{2})^{n-1}.$$

We need

$$t_n = 128.$$

Hence,

$$\begin{aligned} 2(\sqrt{2})^{n-1} &= 128 \\ (\sqrt{2})^{n-1} &= 64. \end{aligned}$$

Since

$$64 = (\sqrt{2})^{12},$$

we get

$$\begin{aligned} n - 1 &= 12 \\ n &= 13. \end{aligned}$$

Final Answer

128 is the 13th term.

Question 7

The Sierpiński square carpet is constructed by repeatedly dividing each square into 9 equal smaller squares and removing the centre square.

Consider Stages 0, 1, 2, 3, . . .

- (i) Find the number of red squares in Stages 0 to 3.
- (ii) Predict the number of red squares in Stages 4 and 5.
- (iii) Find the explicit and recursive formulas for the number of red squares at Stage n .
- (iv) If the area of the original square is 1 square unit, find the area of the red region at Stages 1 to 5. Also find explicit and recursive formulas for the area at Stage n .

Solution**(i) Number of red squares**

At every stage, each red square produces 8 new red squares.

Therefore,

$$\text{Stage } 0 = 1,$$

$$\text{Stage } 1 = 8,$$

$$\text{Stage } 2 = 8^2 = 64,$$

$$\text{Stage } 3 = 8^3 = 512.$$

Thus,

$$\boxed{1, 8, 64, 512}$$

(ii) Stages 4 and 5

$$\begin{aligned}\text{Stage } 4 &= 8^4 \\ &= \boxed{4096},\end{aligned}$$

and

$$\begin{aligned}\text{Stage } 5 &= 8^5 \\ &= \boxed{32768}.\end{aligned}$$

(iii) Explicit and recursive formulas

The sequence is

$$1, 8, 64, 512, \dots$$

Hence,

$$\boxed{R_n = 8^n}$$

is the explicit formula.

For the recursive formula,

$$\boxed{R_0 = 1, \quad R_n = 8R_{n-1}, \quad n \geq 1}$$

(iv) Area of the red region

At each stage, the square is divided into 9 equal parts and 1 part is removed.

Therefore, $\frac{8}{9}$ of the previous area remains.

So the areas are

Final Answer

$$R_n = 8^n$$

$$R_0 = 1, \quad R_n = 8R_{n-1}$$

$$A_n = \left(\frac{8}{9}\right)^n$$

$$A_0 = 1, \quad A_n = \frac{8}{9}A_{n-1}$$

Remember**GP Quick Revision**

$$\text{GP} \rightarrow \text{Same Ratio}$$

$$r = \frac{\text{next term}}{\text{previous term}}$$

$$t_n = ar^{n-1}$$

$$t_1 = a, \quad t_{n+1} = rt_n$$

Easy Trick:

AP $\rightarrow$ Add / Subtract

GP $\rightarrow$ Multiply / Divide

“Find the ratio, understand the pattern, and GP becomes easy!”

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